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Review II Project Phase – II (ECL702)

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1. Title of the Project :

WorkPulse - AI-Powered Multi-Tenant Team Management Platform

2. Introduction :

This project involves the design and development of a web-based team management platform powered by Artificial Intelligence. The application is built using the MERN stack and follows a multi-tenant architecture, allowing multiple organizations to use the same application while keeping their data completely isolated. The system enables companies to manage teams, assign tasks, monitor progress, and generate insights. It also includes AI-based features such as PDF summarization, automated analysis of Excel files, and graphical representation of data using charts and pie diagrams. The AI assistant helps organizations understand productivity trends and improve workflow efficiency.

3. Problem Definition :

Many organizations currently rely on multiple independent tools for managing teams, analyzing documents, and generating reports. This fragmented approach leads to inefficient workflows, difficulty in tracking progress, and lack of centralized insights for decision-making. As a result, productivity decreases and organizations struggle to manage information effectively.

Traditional project and team management systems mainly focus on task assignment and collaboration but often lack advanced capabilities such as AI-driven analytics, automated document summarization, and intelligent data visualization. Additionally, most existing solutions are not designed to efficiently support multiple organizations within a single secure platform, which limits scalability.

Therefore, there is a need for a unified and intelligent platform that can integrate team management, AI-based data analysis, and reporting features into one system. Such a system should:

- Provide centralized team and task management
- Use Artificial Intelligence to generate automated insights
- Summarize documents such as PDFs and reports
- Analyze and visualize data from spreadsheets
- Support multiple organizations securely using a multi-tenant architecture

This project aims to address these challenges by developing a scalable web-based platform using the MERN stack (MongoDB, Express.js, React.js, Node.js) integrated with AI capabilities. The proposed system will improve productivity, provide intelligent insights, and enable organizations to manage their teams and data efficiently within a single platform.

4. Objectives :

- To design and develop a multi-tenant SaaS platform using the MERN stack with secure user authentication and company-based data isolation.
- To implement team and task management features along with AI-powered functionalities such as PDF summarization and a productivity assistant for workflow insights.
- To analyze Excel data for generating charts and visual reports, and deploy the platform on a cloud environment.

5. Skillset additionally required to solve/address the problem :

- JavaScript (ES6+) programming
- React.js for frontend development
- Node.js and Express.js for backend APIs
- MongoDB and Mongoose for database management
- JWT authentication and role-based authorization
- REST API design and integration
- Multi-tenant system architecture
- Tailwind CSS for UI styling
- File handling (PDF and Excel parsing)
- Recharts for data visualization (charts and pie graphs)
- AI API integration
- Cloud deployment (Vercel for frontend, Render for backend)

6. Timeline to achieve the skillset :

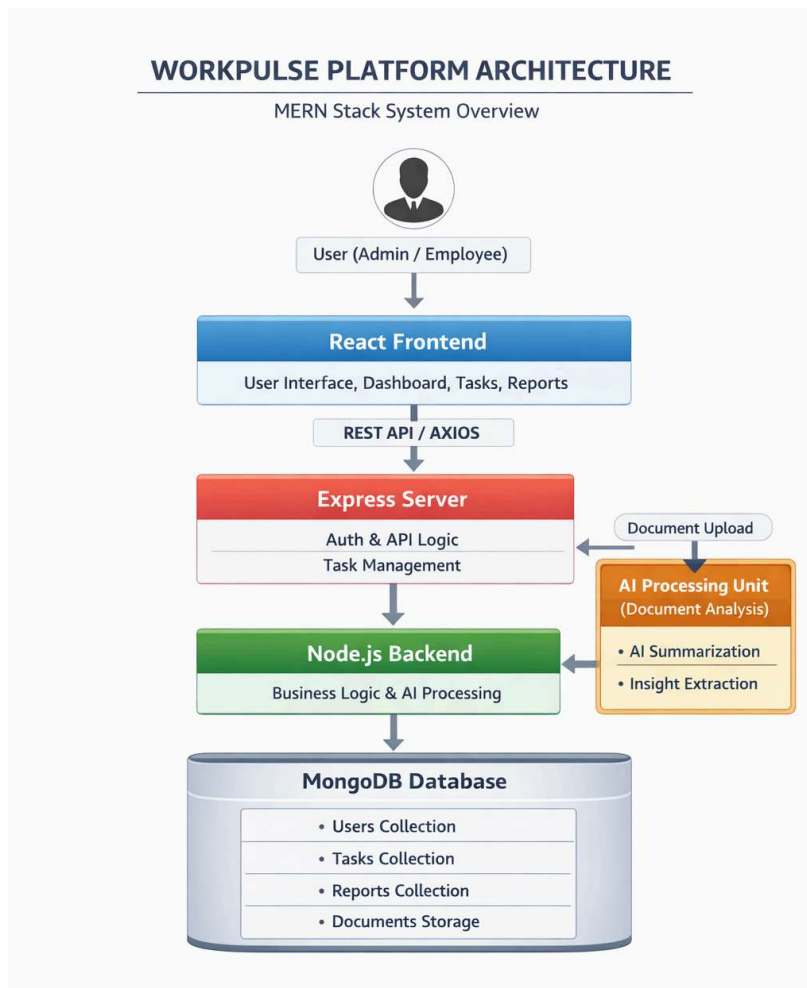
Skillset / Module	Timeframe (Weeks)	Activities to be Performed
MERN Stack Setup & Authentication	1–2	Initialize React frontend and Node.js/Express backend, configure MongoDB with Mongoose, implement JWT authentication, build user registration and login system, and establish multi-tenant architecture with company creation and unique company code generation.
Team & Task Management	3–4	Develop APIs for Teams and Tasks with company-level data isolation, implement task creation and assignment workflows, build Kanban board interface with Todo / In Progress / Done columns, and create a dashboard with task statistics and progress tracking.
Company Join System & Role Management	5–6	Implement company join using company codes, create Users module for viewing organization members, integrate role-based access (Admin / Member), restrict admin-only features such as team creation and user management, and improve UI with consistent dark theme using Tailwind CSS.
Collaboration & Security Enhancements	7–8	Implement role-based backend authorization middleware, develop activity logs for team actions, add task comments and collaboration features, implement "My Tasks" filter view for members, and integrate WebSockets for real-time task updates across users.
AI Document Processing	9–10	Implement PDF upload functionality, integrate API for automatic PDF summarization, and display generated summaries within the dashboard for quick insights.
Excel Analysis & Data Visualization	11–12	Develop backend parsing for Excel files, generate analytics from uploaded data, and integrate Recharts library to display visual insights such as bar charts and pie diagrams on the dashboard.

7. Block schematic of the Project:

The block schematic diagram of WorkPulse illustrates the overall architecture of the system developed using the MERN stack (MongoDB, Express.js, React.js, Node.js). The user interacts with the system through the React frontend, which provides features such as task management, dashboards, and document uploads.

The frontend communicates with the Express/Node.js backend using REST APIs. The backend processes requests, manages authentication, and performs application logic including AI-based document analysis.

All application data such as users, tasks, reports, and uploaded documents are stored in the MongoDB database. This architecture enables efficient data handling, centralized team management, and automated insights within the WorkPulse platform.



8. Weekly milestones:

Week	Milestone
Week 1	Initialized React frontend with Tailwind CSS, React Router, and Node.js/Express backend with modular folder structure
Week 2	Connected MongoDB with Mongoose, implemented JWT authentication, built Register and Login pages with multi-tenant company code generation
Week 3	Defined Team and Task schemas, built REST APIs with company-level data isolation, implemented task assignment to members
Week 4	Developed Kanban board UI (Todo / In Progress / Done), built Dashboard with stats cards and company code copy button
Week 5	Implement company join via code, build Users module, integrate role-based access control (Admin / Member)
Week 6	Restrict admin-only pages, apply AdminRoute and ProtectedRoute across all frontend routes
Week 7	Add role-based backend authorization middleware and implement activity logs for team actions
Week 8	Add task comments, build "My Tasks" filter for members, integrate WebSockets for real-time updates
Week 9	Build PDF upload functionality on frontend and backend
Week 10	Integrate API for PDF summarization and display results on dashboard
Week 11	Develop backend Excel parsing and generate analytics from uploaded data
Week 12	Integrate Recharts for bar charts and pie diagrams on a dedicated analytics page

9. Completed Milestones :

- **Project Setup:** Initialized React frontend and Node.js/Express backend with folder structure and environment configuration.
- **Database Configuration:** Connected MongoDB with Mongoose and defined schemas for User, Company, Team, and Task models.
- **Authentication System:** Implemented JWT-based registration and login with secure password hashing.

- **Multi-Tenant Architecture:** Built company creation with unique auto-generated company codes and company-based data isolation across all modules.
- **Company Join System:** Enabled new users to join an existing company using a company code during registration.
- **Role-Based Access Control:** Implemented Admin and Member roles with protected frontend routes (AdminRoute, ProtectedRoute) and role-restricted sidebar navigation.
- **Team Management:** Built API and UI for creating and viewing teams scoped to the user's company.
- **Task Management:** Developed full task creation, assignment, and status update system with a Kanban board (Todo / In Progress / Done).
- **Dashboard:** Built statistics dashboard showing total teams, tasks, completed and in-progress counts, company code display with copy button, and members list.
- **Users Module:** Created Users page for admins to view all members of their organization with role badges.
- **My Tasks Filter:** Added a personalized task view for members showing only their assigned tasks.

10. Milestones to be Completed:

- **Company Join via Invite Link:** Generate unique invite links for companies, allow users to register and join a company by clicking the invite link instead of manually entering a code.
- **Role-Based Access Control:** Restrict Teams and Users pages to admin only using AdminRoute and ProtectedRoute components.
- **Backend Authorization Middleware:** Implement server-side role checks to protect admin-only API endpoints.
- **Activity Logs:** Track and display team actions such as task creation, updates, and assignments.
- **Task Comments & Collaboration:** Allow team members to comment on tasks for better communication.
- **Real-Time Updates:** Integrate WebSockets to sync task status changes instantly across all connected users.
- **AI PDF Summarization:** Build PDF upload and integrate API to automatically generate document summaries.
- **Excel File Analysis:** Develop backend Excel parsing to extract and process data from uploaded files.
- **Data Visualization:** Integrate Recharts to display bar charts and pie diagrams from analyzed Excel data.
- **Cloud Deployment:** Deploy frontend to Vercel and backend to Render with MongoDB Atlas as the production database.

11. Expected Challenges:

- **Invite Link Security:** Generating tamper-proof, expiring invite links and handling edge cases such as expired links, already-used links, and unauthorized access attempts.
- **Role-Based Access Consistency:** Maintaining role enforcement consistently across both frontend route protection and backend API middleware to prevent privilege escalation.
- **Real-Time Synchronization:** Managing WebSocket connections efficiently across multiple users and companies without performance degradation or data conflicts.
- **AI API Integration:** Handling large PDF documents within token limits of the API, managing API rate limits, and ensuring accurate and meaningful summaries.
- **Excel Data Variability:** Parsing Excel files with inconsistent formats, missing values, or varied column structures while still generating meaningful charts and analytics.
- **File Upload Handling:** Managing large file uploads securely on the backend, validating file types, and storing or processing files without impacting server performance.
- **Chart Rendering Performance:** Rendering large datasets in Recharts without causing UI lag or memory issues, especially on the analytics dashboard.
- **JWT Token Management:** Handling token expiry, refresh cycles, and secure storage on the client side to maintain seamless and secure user sessions.

12. References :

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Name and Signature of the Student

Name and Signature of Supervisor